Risk-Based Segmentation for Financial Asset Recommendation: Balancing Return, Accuracy, and Risk Suitability

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Abstract

We present risk-based segmentation, a financial asset recommendation (FAR) approach that routes each customer into a per-band sub-model of LightGCN, so their declared risk band under the Markets in Financial Instruments Directive II (MiFID II) shapes recommendations during training. On top of this segmentation we ask where suitability is best enforced, comparing two exclusive interventions: steering the model with a coherence margin loss, or making the data coherent by regrouping each customer onto the band implied by their purchases. This places risk suitability inside the model, whereas existing FAR systems optimise only return and ranking accuracy or correct for risk by post-hoc re-ranking. We evaluate on three axes: return (ROI), ranking accuracy (nDCG), and risk suitability, which we make measurable with profile coherence (PC), a metric we introduce, and summarise with their harmonic mean as a single balance. An analytical study of FAR-Trans [Sanz-Cruzado *et al.*, 2024] motivates this third axis: 18.6% of scoreable buys lie two or more bands from the customer's declared tolerance, a persistent customer-level trait, and a regression shows coherent buys earn $+2.94$ percentage points higher 6-month realised return than discordant ones. Across 69 temporal splits, the margin loss lifts coherence but sacrifices accuracy, whereas regrouping lifts it while also improving nDCG and leaving ROI unchanged, the only intervention to raise the overall balance. Paired with regrouping, risk-based segmentation provides the strongest gains and offers a template for suitability-constrained recommendation in other regulated domains.

1 Introduction

Financial asset recommendation (FAR) identifies and ranks financial securities such as stocks, bonds, and mutual funds for investors according to their suitability. It draws on heterogeneous signals, combining a customer's past transactions and declared risk tolerance on the investor side with asset prices, returns, and volatility on the market side. FAR therefore sits at the intersection of personalised ranking and regulatory obligation. The Markets in Financial Instruments Directive II (MiFID II) is a foundational European Union regulation that requires firms to assess suitability before recommending a product, so that the product's risk aligns with the customer's declared risk tolerance and loss capacity. A recommender that maximises predicted return or ranking quality without conditioning on this declared risk band can excel on standard benchmarks yet remain a regulatory liability.

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first open FAR benchmark to pair real asset pricing with real retail-investor transactions and a declared MiFID II risk band per customer, collected from a large European financial institution over January 2018 to November 2022. Its accompanying benchmark evaluates FAR algorithms on two axes: ranking accuracy, via normalised Discounted Cumulative Gain (nDCG@10), and realised profitability, via Return on Investment (ROI@10). A subsequent study finds these two axes are themselves negatively correlated on FAR-Trans, so a model that wins on accuracy tends to lose on return [Sanz-Cruzado *et al.*, 2026].

Neither axis measures whether a recommendation is suitable for the customer's risk band. A model can lead the nDCG or ROI leaderboard while routinely proposing assets two or more bands away from the customer's declared tolerance, the precise failure MiFID II exists to prevent. Risk-aware FAR methods address this only downstream, re-ranking a base model's top list by a risk-tolerance utility [Sakurai *et al.*, 2026] rather than conditioning the model on the band. The declared risk band therefore never enters training, and risk suitability is left to a post-hoc filter or ignored outright.

Our objective is to balance risk suitability, the alignment between an asset's risk and the customer's declared band, against return and accuracy rather than trade one for another. We pursue this through a risk-based segmentation of the FAR model, then experiment with two interventions on top of it, a coherence margin loss and customer regrouping. We evaluate the models on three axes: return (ROI@10), ranking accuracy (nDCG@10), and risk suitability, which we make measurable by introducing Profile Coherence at k (PC@k). PC@k is the share of a customer's top-k list whose asset risk class lies within one band of the customer's declared MiFID II band, on a four-point ordinal scale from Conservative to Aggressive. To capture how well a recommender reconciles the three, we further introduce a single balance score, their harmonic mean, which is high only when return, accuracy, and risk suitability are strong together and falls sharply when any one is sacrificed.

Concretely, the segmentation builds on LightGCN [He *et al.*, 2020]: we partition customers by declared band and route each into a dedicated sub-model during training, so the band shapes the learned embeddings directly rather than filtering its output. The margin loss extends Bayesian Personalised Ranking (BPR) [Rendle *et al.*, 2012] to penalise scoring a discordant asset above a coherent one, whereas regrouping instead corrects the data by reassigning each customer to the band implied by their revealed purchases. In summary, we make three contributions:

- **An analytical study of risk factors in FAR-Trans.** We establish risk suitability as a measurable axis, showing that discordance between a customer's declared band and the assets they buy is widespread and a persistent customer-level trait, and that coherent buys earn a return premium over discordant ones rather than a penalty.
- **A risk-based segmentation of LightGCN.** The baseline methodology that conditions training on the declared risk band, on which the two interventions build.
- **Experimental studies across the three axes.** Across 69 temporal splits we audit the FAR-Trans baseline models and ablate the two exclusive interventions, finding that the baselines under-serve specific risk bands and that only regrouping the data raises the harmonic-mean balance, by lifting suitability and accuracy at no cost to return.

2 The FAR-Trans Benchmark

Most prior FAR models were developed over proprietary or simulated data, which prevents fair cross-method comparison, and the only earlier public dataset lacked pricing data and asset identifiers, ruling out price-based approaches. FAR-Trans [Sanz-Cruzado *et al.*, 2024] closes this gap. It provides daily closing prices, 228,913 buy and 159,135 sell transactions over 806 assets and 29,090 predominantly retail customers, and customer profile metadata including a declared MiFID II risk band per customer. Together these make both price-based and profile-based recommendation measurable on a single dataset.

Alongside the data, FAR-Trans establishes the field's evaluation standard by benchmarking 11 FAR algorithms. Two anchor our study because each leads on one axis of that benchmark: LightGCN [He *et al.*, 2020], a graph-based collaborative filter that propagates embeddings over the user-item bipartite graph and is the strongest ranking model, and Random Forest, a tree-based ensemble that ranks assets from tabular customer and asset features and is the strongest profitability model. The benchmark scores both over the top-10 ranked items on two axes: normalised Discounted Cumulative Gain (nDCG@10) for next-purchase ranking accuracy and Return on Investment (ROI@10) for realised profitability.

3 Analytical Study of Risk Factors

Measuring risk suitability requires customers and assets on a common scale. FAR-Trans supplies a declared MiFID II risk band per customer but none for assets, so we map each asset onto the same ordinal scale with a hierarchical rule of our own. Profile Coherence then scores how closely a recommendation matches the customer's band.

3.1 Risk Bands

Ordinal band scale

Customers and assets share the same 4-point ordinal scale running from Conservative to Income to Balanced to Aggressive, encoded 0 to 3. The customer band b_u is taken from the customer's declared risk level in FAR-Trans. FAR-Trans provides both declared and regression-imputed risk levels, and both are mapped to the same ordinal scale. Customers whose risk level is unavailable or missing are excluded from all coherence computations. The asset band b_i is assigned by the three-step hierarchical rule below, applied in order.

Step 1: subcategory metadata

Mutual funds are labelled directly from their subcategory: Money Market funds are treated as Conservative, Bond funds as Income, Balanced as Balanced, and Equity or Large Cap as Aggressive. Bond instruments are labelled by issuer type: Government as Conservative, Corporate as Income, and any remaining bond subcategory as Income by default.

Step 2: volatility quartiles for stocks

Stocks carry no subcategory tag we can use, so they fall through to a volatility rule. For each ISIN we compute the annualised standard deviation of daily log returns on a trailing 252-day window, then bucket all stocks into the four bands by the stock-only quartile thresholds (q_1, q_2, q_3).

Step 3: residual default

Any asset that falls through both prior steps is assigned to Balanced.

Applying the three steps above produces an asset-band distribution of $(190, 333, 105, 178)$ across (Conservative, Income, Balanced, Aggressive) over a total of 806 assets.

3.2 Profile Coherence

Risk suitability is the alignment between an asset's risk band and the customer's. We make it measurable with Profile Coherence, built from the ordinal distance between the two bands.

Pairwise discordance

$$d(u, i) = |b_u - b_i| \in \{0, 1, 2, 3\}. \tag{1}$$

Here u denotes a customer and i an asset, $b_u \in \{0, 1, 2, 3\}$ is the customer's ordinal risk band and $b_i \in \{0, 1, 2, 3\}$ the asset's risk classification, so the discordance $d(u, i)$ is the absolute gap between the two bands. A recommendation is *profile-coherent* iff $d \leq 1$. We adopt this ± 1 band tolerance by design, to mirror MiFID II's adjacency allowance for borderline suitability rather than to penalise every single-band drift as a violation.

Aggregation

$$\text{PC@}k(u) = \frac{1}{k} \left| \{ i \in \text{top}_k(u) : d(u, i) \leq 1 \} \right|. \tag{2}$$

Here k is the rank cutoff, $\text{top}_k(u)$ is the set of the k highest-scored assets recommended to customer u, and $|\cdot|$ counts how many of them are coherent ($d(u, i) \leq 1$), so PC@$k(u)$

is the fraction of u's top-k recommendations that are profile-coherent. Two implementation rules complete the definition. First, recommenders returning fewer than k items use the actual number of items returned as the denominator. Second, customers with no assigned risk band ($b_u = \emptyset$) are dropped from the aggregate. Per-split PC@k is the unweighted mean across eligible customers and the headline numerics average over all splits.

3.3 Dataset

Beyond the dataset scale given in Section 2, two profile details matter for coherence. Of the 29,090 customers, 28,770 (98.9%) carry a declared or regression-imputed MiFID II risk band, and the remaining 320 are excluded from all coherence computations. Customer profiles also include the type of customer, which enters our regression study later as a control.

3.4 Prevalence and Structure of Discordance

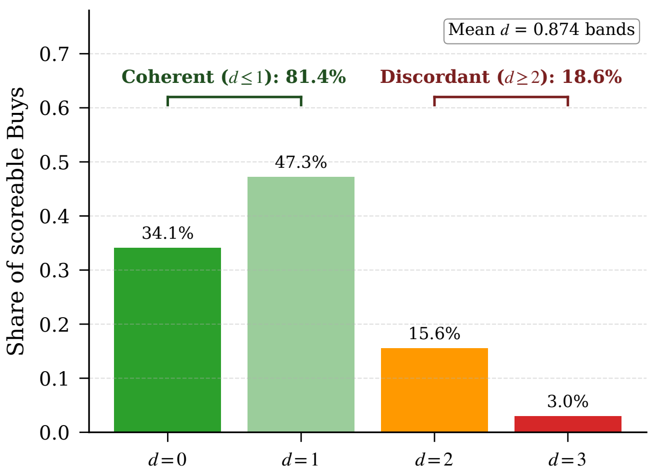

Figure 1: Distribution of pairwise discordance d over 228,241 scoreable buys. Most mass sits at $d \leq 1$ (coherent), with non-negligible density at $d \geq 2$ (discordant).

At the transaction level (Figure 1), of 228,241 scoreable buys:

1. 18.6% violate the declared band by two or more steps.
2. 81.4% are coherent, and 34.1% are strictly coherent.
3. Mean discordance is 0.874 bands across all scoreable buys, indicating that when mismatches occur they are usually around a single band step.

All in all, nearly one in five scoreable buys lands two or more bands away from the customer's declared tolerance. That violation rate is not small to dismiss as sampling noise, and it is the empirical motivation for treating risk suitability as a measurable evaluation axis rather than a property the baselines can be assumed to satisfy.

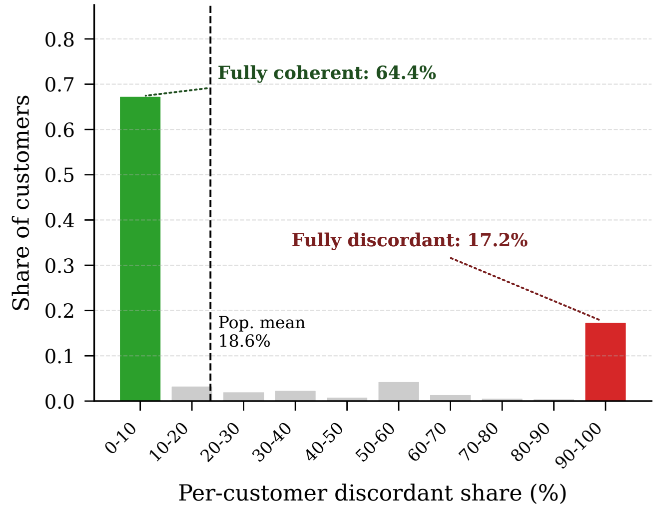

Figure 2: Per-customer share of discordant buys is sharply bimodal, concentrating in the lowest and highest bins and ruling out i.i.d. transaction-level noise.

The pooled rate cannot tell us whether these violations are i.i.d. transaction-level slips spread thinly across the customer base or a structural trait concentrated in a subset. Slicing by customer settles the question. The per-customer discordant share (Figure 2) is sharply bimodal, concentrating in the lowest and highest bins rather than clustering near the 18.6% population mean. More explicitly, 64.4% of banded customers are fully coherent across every buy and 17.2% are fully discordant.

Hartigan's dip test on the 28,770 per-customer values rejects unimodality ($D = 0.086$, $p < 10^{-15}$), confirming that discordance is a persistent customer-level trait rather than i.i.d. transaction-level noise.

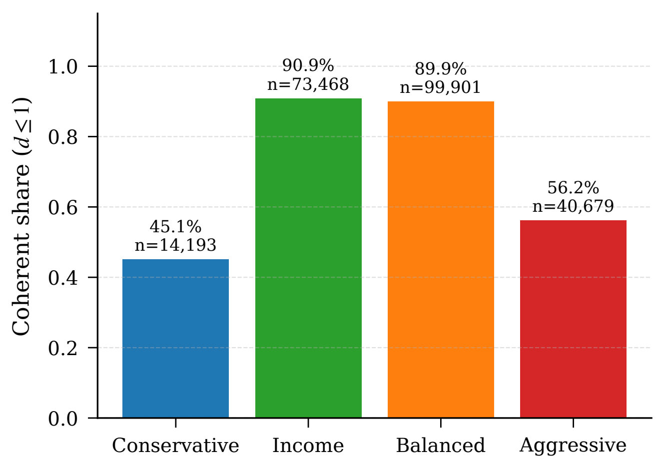

Figure 3: Coherence share by declared band. The U-shape supports the bimodality result of Figure 2: discordance concentrates on the ordinal extremes.

Since the trait is customer-level, the next question is which customers carry it. Coherence per declared band is U-shaped (Figure 3). The two centre bands sit around 90%, while coherence falls to 45.1% for Conservative customers and 56.2%

for Aggressive ones. The Conservative dip is consistent with a reach for yield, where customers buy assets riskier than their declared band, and the Aggressive dip with a pull towards safer assets than their band implies. This concentration of mismatch at the ordinal extremes corroborates the bimodality finding from the customer slice.

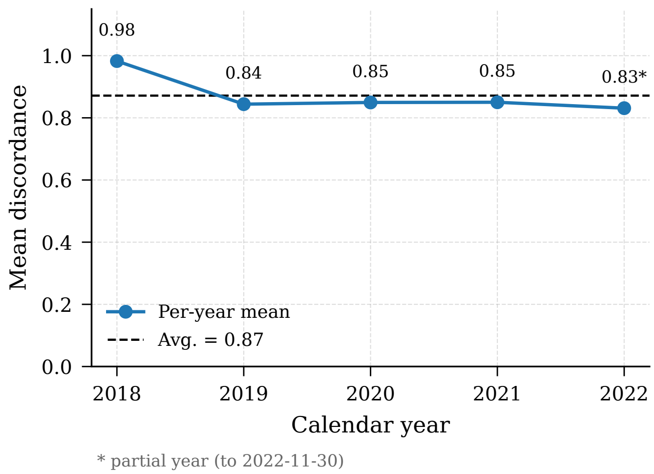

Figure 4: Mean discordance by calendar year. The 2019 to 2022 cluster sits within 0.02 bands, with 2018 elevated (transition-effect hypothesis).

A final slice rules out a regime-driven artefact. Mean discordance is flat across 2019 to 2022, all within 0.02 bands (Figure 4), so the pattern is not the product of any single year's market conditions. The 2018 value of 0.98 sits higher. MiFID II came into force on 2018-01-03, the exact start of FAR-Trans, so we hypothesise that the elevation is a transition effect. However, this hypothesis is not tested here and is left as possible future work.

3.5 Conditional Return Premium

We have shown that discordance is a persistent customer-level trait concentrated at the ordinal extremes. The next question is whether it carries an economic cost: do discordant buys earn lower returns than coherent ones?

Regression specification

We test whether profile-coherent buys earn different realised returns from discordant ones via a transaction-level Ordinary Least Squares (OLS) regression on 208,029 priced buys with cluster-robust standard errors at the customer level and a two-sided Wald test at $\alpha = 0.05$. Clustering is warranted because each customer contributes roughly 8 buys on average, so within-customer residuals share unobserved shocks. The specification is:

$$\begin{aligned} r = \beta_0 &+ \beta_1 \cdot \text{is\_coherent} \\ &+ \beta_2 \cdot \text{asset\_volatility} \\ &+ \mathbf{C}(\text{type})\boldsymbol{\gamma} \\ &+ \mathbf{C}(\text{year})\boldsymbol{\delta} \\ &+ \varepsilon, \end{aligned} \tag{3}$$

where $r = (p_{\text{end}} - p_{\text{start}})/p_{\text{start}}$ is the 6-month realised return as a decimal fraction (e.g., $r = 0.03$ means a 3 percentage-point gain) and the coherence indicator is binary ($d \leq 1$).

Prices are matched via a backward-nearest join: the start price is the most recent close on or before the trade date, and the end price is the most recent close on or before the date six months later. Transactions without a price match within 7 days at either endpoint are dropped. The coefficient of interest is β_1 on the coherence indicator, and the null hypothesis $H_0\colon \beta_1 = 0$ states that coherence has no conditional effect on 6-month returns.

Regression Results

Table 1 reports the full coefficient table. The key results on the coherence premium are:

- **Conditional premium.** Coherent buys earn $+2.94$ percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year.
- **Significance.** With $p = 2.3 \times 10^{-13}$, we reject the null hypothesis $H_0\colon \beta_1 = 0$ at $\alpha = 0.05$. This rejection does not prove a causal effect, but the significantly positive estimate gives no support to the view that coherence has no conditional effect on returns.

Turning to the controls, the volatility coefficient is -10.25 pp per unit, consistent with low-volatility outperformance over 2019 to 2022 and the 2022 rate shock that hit high volatility products. Each of the four year dummies is individually significant, rejecting its own null hypothesis and reflecting distinct market regimes relative to the 2018 reference year:

1. The 2019 coefficient reflects the broad equity recovery in the year after the December 2018 sell-off, which had left the S&P 500 just short of bear-market territory [CNBC, 2019].
2. 2020 is the strongest at $+21.07$ pp, coinciding with the rapid rebound from the COVID-19 sell-off that returned major indices to record highs by August [Reuters, 2020].
3. 2021 reflects continued gains under still-accommodative monetary policy, before the Federal Reserve's expected late-2021 taper and rising rate-hike expectations [Reuters, 2021].
4. The 2022 coefficient turns negative at -1.95 pp, the year an aggressive Federal Reserve hiking cycle drove a broad fixed-income drawdown [Reuters, 2022].

Lastly, none of the four customer-type dummies is significant, each failing to reject its own null hypothesis, indicating that segmentation of customer type contributes no detectable return signal. Together with the bimodality found above, this suggests that customer type does not capture the persistent discordance trait either.

Taken all together, these controls barely move the premium: the raw slice gap is $+3.31$ pp (unconditional mean 6-month return: 4.23% coherent vs 0.92% discordant), whereas the coherence coefficient, which already conditions on volatility, segment, and year, is $+2.94$ pp. Adding the controls therefore absorbs only 0.37 pp which is around 11% of the unconditional gap, so coherence is not a volatility, segment, or year

Term	Estimate	Std. Error	95% CI	p
Intercept	-0.0471	0.0141	$[-0.0746, -0.0195]$	8.1×10^{-4}
`is_coherent`	$+0.0294$	0.0040	$[+0.0215, +0.0372]$	2.3×10^{-13}
`asset_volatility`	-0.1025	0.0165	$[-0.1349, -0.0701]$	5.5×10^{-10}
`C(customer_type)[Legal Entity]`	$+0.0154$	0.0486	$[-0.0798, +0.1106]$	0.751
`C(customer_type)[Mass]`	$+0.0011$	0.0132	$[-0.0248, +0.0269]$	0.936
`C(customer_type)[Premium]`	$+0.0058$	0.0135	$[-0.0206, +0.0322]$	0.667
`C(customer_type)[Professional]`	$+0.0247$	0.0169	$[-0.0084, +0.0578]$	0.143
`C(year)[2019]`	$+0.1759$	0.0129	$[+0.1507, +0.2012]$	1.4×10^{-42}
`C(year)[2020]`	$+0.2107$	0.0061	$[+0.1986, +0.2227]$	6.2×10^{-259}
`C(year)[2021]`	$+0.0754$	0.0053	$[+0.0651, +0.0858]$	4.1×10^{-46}
`C(year)[2022]`	-0.0195	0.0045	$[-0.0283, -0.0107]$	1.4×10^{-5}

Table 1: Full OLS coefficient table for the return-premium regression on 208,029 priced buys, cluster-robust SE on Customer ID.

confound. The coherence premium therefore survives all controls we can measure, motivating the question of whether existing recommenders produce coherent asset lists at all.

3.6 Limitations

Two caveats bound our analytic studies.

- *Observational nature.* The coherence premium is associational, as unobserved confounders such as financial literacy, advisor channel, account constraints, and prior portfolio composition are not ruled out by the volatility, segment, or year controls. Similarly, the elevated 2018 mean discordance is hypothesised as a MiFID II transition effect but not directly tested against a pre-MiFID II counterfactual.
- *Asset-band assignment.* The asset bands b_i are not provided by FAR-Trans but assigned by the hierarchical heuristic in Section 3.1 rather than an externally validated classification, so any systematic mislabelling, especially from the volatility-quartile step for stocks, propagates into all coherence numbers.

4 Methodology: Risk-Based Segmentation

Our methodology brings risk suitability inside the recommender. The core idea is *risk-based segmentation*: instead of a single global model, we route each customer to a sub-model trained only on customers of their risk band. Figure 5 gives a full picture of our methodology.

4.1 Suitability Gap of Single Global Model

The FAR-Trans LightGCN baseline is a single collaborative filter over one user-item bipartite graph that connects every customer to the assets they bought. It learns an embedding per customer and per asset, propagates them over the graph, and scores an asset for a customer by the inner product of their embeddings, training with the Bayesian Personalised Ranking (BPR) objective [Rendle *et al.*, 2012]. The model never sees the customer's risk band, so suitability is whatever the purchase graph happens to imply. Section 3 showed that purchases are frequently discordant, a band-agnostic model trained to reproduce them inherits that discordance.

4.2 Risk-Based Segmentation

We choose LightGCN as the backbone because its differentiable loss and per-customer embeddings let us condition on the risk band and, later, attach a coherence margin directly to the BPR objective. Prior work shows LightGCN readily carries domain-specific structure beyond plain BPR [Ghiye *et al.*, 2024]. Random Forest, the other FAR-Trans anchor, ranks assets from technical indicators alone and carries no customer-level signal to segment on, so we do not extend it.

Risk-based segmentation partitions the customer base by declared MiFID II band and trains one LightGCN sub-model per band: Conservative, Income, Balanced and Aggressive. Each sub-model's training graph contains only the interactions of customers in that band, so its embeddings specialise to that band's preferences. A band router directs each customer at both training and inference: a customer is served by the sub-model of their declared band, and customers with no declared band fall back to the Balanced sub-model. Each sub-model is trained with the same BPR objective and edge-dropout regularisation as the baseline, changing only the population it learns from.

4.3 Two Alternatives for Raising Risk Suitability

Segmentation specialises each sub-model to a band but still trains purely to reproduce purchases, which Section 3 showed are themselves often discordant. We therefore study two exclusive alternatives for pushing risk suitability higher, one acting on the model and one on the data.

Approach 1: Steering the model via margin loss

BPR trains the model to score a bought asset (the positive) above a random unobserved asset (the negative). Independently of this sampling, we add a coherence margin that draws, from the sub-model's asset universe, one coherent asset i_c ($d(u, i_c) \leq 1$) and one discordant asset i_d ($d(u, i_d) > 1$), and pushes the coherent score above the discordant one:

$$\mathcal{L}_{\text{coh}}(u) = \text{softplus}(s(u, i_d) - s(u, i_c)), \qquad (4)$$

where $s(u, i)$ is the inner-product score and $\text{softplus}(x) = \ln(1 + e^x)$ smoothly approximates the hinge $\max(0, x)$. The total objective adds this term to BPR with weight $\lambda \geq 0$:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \omega\|\Theta\|_2^2 + \lambda \cdot \mathcal{L}_{\text{coh}}, \qquad (5)$$

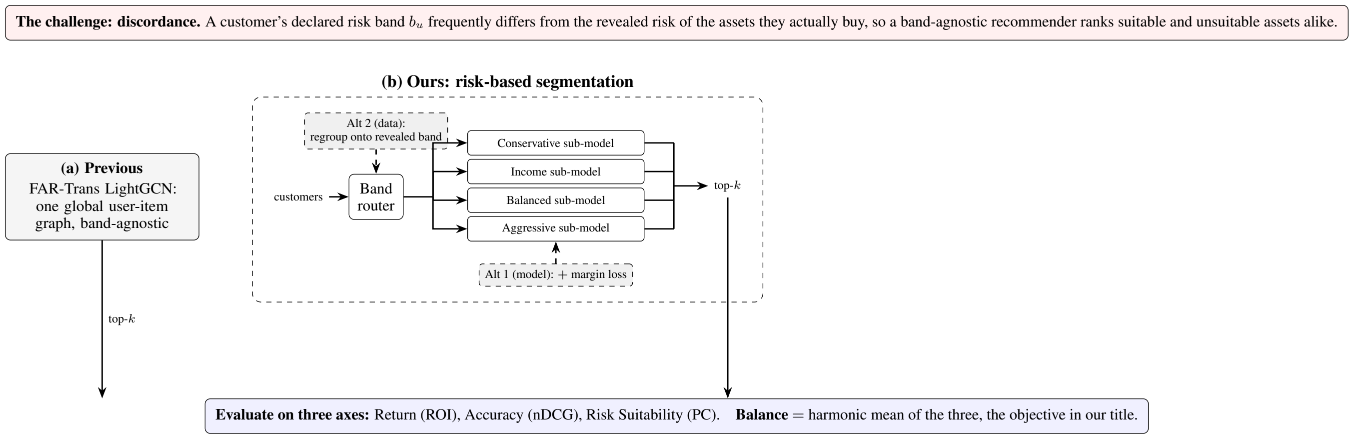

Figure 5: Overview of risk-based segmentation.

where ω is the L2 weight decay on embedding parameters Θ. BPR remains the primary ranking signal: setting $\lambda = 0$ recovers plain segmentation, while $\lambda > 0$ steers scores towards coherent assets. We use softplus rather than a hard hinge because it is differentiable everywhere and is the same form BPR itself uses, keeping the two objectives consistent.

Approach 2: Fixing the data via regrouping

The second approach corrects the label at its source: before training, each banded customer is reassigned to the band implied by their purchases, taken as the lower median of the risk classes of the assets they bought (even-count ties resolve to the more conservative band). Segmentation then routes and trains on these regrouped bands, making the data coherent in the first place rather than penalising incoherence during training.

5 Experimental Setup

5.1 Temporal Split Protocol

Evaluation uses 69 temporal splits with approximately biweekly split points on an evenly spaced trading-day grid running from August 2019 to December 2021, with per-split test windows extending evaluation coverage to May 2022. The two-week split cadence follows the original FAR-Trans benchmark [Sanz-Cruzado *et al.*, 2024]. For each split, the training set contains all buys up to the split point and the test set contains all buys from the split point to the end of the test window, deduplicated against training and filtered to the training-asset universe. Eligible customers have at least one interaction in both training and test, while eligible assets have prices on both endpoints.

5.2 Evaluation Metrics

We score every recommender over its top-10 ranked items on the three axes, then summarise them with a single balance.

Return: ROI@10. For each recommended asset we form the holding-period return from its close price at the split point and at the test-window end, convert it to a per-month geometric rate, average over the ten assets, and then over eligible customers. ROI@10 is reported per month.

Accuracy: nDCG@10. Ranking accuracy is the rank-discounted hit rate of the held-out test buys within the top 10, normalised by the ideal ordering and averaged over eligible customers.

Risk suitability: PC@10. Profile Coherence is defined in Section 3.2. To read per-band coherence on a common footing, we normalise it by a band-conditional random baseline. The closed-form expectation of PC@k under a uniformly random recommender for band b is

$$\pi(b) = \frac{|\{i \in A : |b - b_i| \leq 1\}|}{|A|}, \tag{6}$$

where A is the asset universe. With the asset-band distribution of Section 3.1 this yields $\pi(b) = (0.65, 0.78, 0.76, 0.35)$ for (Conservative, Income, Balanced, Aggressive). Per-user coherence is then normalised against the random reference at the customer's own band,

$$\text{PC-lift@}k(u) = \frac{\text{PC@}k(u)}{\pi(b_u)}, \tag{7}$$

so that lift > 1 beats the band-conditional random rate. Because 77.9% of FAR-Trans assets sit in the Conservative, Income, and Balanced bands, a recommender that recommends Income assets to everyone would score well on raw PC@k for any customer whose tolerance set includes Income. Dividing by $\pi(b_u)$ removes this advantage, so bands that are easier to satisfy by chance are not credited for it and the lift values are comparable across bands.

Harmonic Balance. We summarise the three axes with a single balance score, their harmonic mean. nDCG@10 and PC@10 are already rates in $[0, 1]$, but ROI@10 can be negative and lives on a different scale, so we first map it onto $[0, 1]$ with a logistic transform

$$\text{ROI}' = \frac{1}{1 + e^{-\text{ROI@10}/\tau}}, \qquad \tau = 0.01, \tag{8}$$

which sends a flat return to 0.5 and moves smoothly towards 1 for gains and 0 for losses, with τ setting the scale at one percentage point per month. The balance is then

$$\text{Balance} = 3 \left(\frac{1}{\text{ROI}'} + \frac{1}{\text{nDCG@10}} + \frac{1}{\text{PC@10}} \right)^{-1}, \tag{9}$$

which is high only when return, accuracy, and risk suitability are strong together and collapses if any one is sacrificed.

5.3 Baselines and Hyperparameter Grids

Parameter	Search space	Best	Role
LightGCN (tuned on average nDCG@10)			
`embedding_dim`	$\{64, 128\}$	64	grid
`num_layers`	$\{2, 3\}$	3	grid
`learning_rate`	$\{10^{-2}, 10^{-3}\}$	10^{-3}	grid
`weight_decay`	10^{-5}	n/a	fixed
`keep_prob`	0.6	n/a	fixed
`num_epochs`	50	n/a	fixed
`batch_size`	1024	n/a	fixed
Random Forest (tuned on average ROI@10)			
`num_estimators`	$\{20, 30, 40, 50\}$	40	grid
`max_depth`	$\{15, 25, 50\}$	50	grid
horizon	6 months	n/a	fixed
`random_state`	42	n/a	fixed

Table 2: Baseline hyperparameter grids and best-trial values, swept with Ray Tune. "grid" rows are searched, "fixed" rows are held constant.

Table 2 reports the search grids, fixed values, and best-trial selections for the two FAR-Trans baselines. Both grids are swept with Ray Tune, with LightGCN tuned on average nDCG@10 and Random Forest on average ROI@10 across the 69 splits.

5.4 Configurations and Per-Band Audit

We compare the following configurations on the same 69 splits, each reported on all three axes and the balance:

- **Baseline.** The FAR-Trans LightGCN at its best trial, and the Random Forest baseline for the per-axis reference.
- **Segmentation.** Risk-based segmentation alone without the coherence margin loss, isolating the contribution of the architecture.
- **Model lever.** Segmentation with the coherence margin loss added.
- **Data lever.** Segmentation retrained and routed on the regrouped bands.

All segmented configurations inherit the best LightGCN backbone hyperparameters of Table 2, and the segmentation is trained with the same BPR objective and edge-dropout regularisation as the baseline.

For the per-band audit, we average per-customer PC@10 unweighted over all (customer, split) pairs within each declared band, and report per-band PC-lift@10 against $\pi(b)$. This exposes which bands a recommender actually serves, which the aggregate PC@10 can mask because the random baselines $\pi(b)$ differ widely across bands.

6 Results and Discussion

Across the three axes, only one intervention improves on the LightGCN baseline overall: regrouping the data. Aggregate metrics over the 69 splits are in Table 3 and per-band suitability in Table 4.

6.1 Baseline Audit

The two FAR-Trans baselines sit at opposite corners of the return-accuracy tradeoff, and neither balances the three axes (Table 3):

- **Random Forest** wins return ($+1.4\%$/mo) but ranks near random (nDCG@10 of 0.019), so its balance collapses to 0.054.
- **LightGCN** wins ranking accuracy (nDCG@10 of 0.330, PC@10 of 0.794) but earns a slightly negative ROI@10, for a balance of 0.428.
- Aggregate PC@10 looks adequate for both ($1.17\times$ lift for LightGCN, $1.06\times$ for Random Forest), but this single number averages over bands whose random baselines $\pi(b)$ span 0.35 to 0.78 and so hides which bands are actually served.

Table 4 exposes that gap. Per-band lift rises monotonically with the declared band for both baselines:

- **Random Forest:** $0.52 \rightarrow 0.70 \rightarrow 1.03 \rightarrow 1.88$ from Conservative to Aggressive.
- **LightGCN:** $0.75 \rightarrow 1.13 \rightarrow 1.17 \rightarrow 1.35$ over the same bands.
- Both skew towards higher-risk assets and under-serve the Conservative band below the random baseline, and Random Forest also under-serves Income.
- Aggressive's high lift is partly mechanical, since $\pi(\text{Aggressive}) = 0.35$ is the lowest bar to clear.

Neither baseline carries band labels or a band-aware objective, so per-band coherence is likely a passive byproduct of training rather than a controlled output. This is the deficit a band-aware approach must close.

6.2 Aggregate Results of Risk-Based Segmentation

Reading the ablation down Table 3 separates a cheap architectural step from the two levers:

- **Segmentation alone** lifts PC@10 to 0.821 at roughly flat ranking (nDCG@10 of 0.332) and a small ROI cost, for a balance of 0.411, marginally below the baseline.
- **Model lever (margin loss)** pushes PC@10 to 0.918 but spends accuracy doing so, dropping nDCG@10 to 0.309, for a balance of 0.407 that also sits below the baseline.
- **Data lever (regrouping)** lifts PC@10 to 0.949 while *also* raising nDCG@10 to 0.352 and leaving ROI@10 essentially unchanged, for a balance of 0.452, the highest of any configuration and the only one above the LightGCN baseline.
- **Combined** reaches the highest raw PC@10 (0.982) but its balance (0.444) falls back below regrouping alone, because the margin loss spends accuracy even on the corrected bands.

Of the two exclusive levers, only regrouping raises the harmonic-mean balance, because it corrects the noisy declared labels at their source rather than fighting them during training. The proof is in nDCG@10 and ROI@10, neither of which uses the risk band: both still rise under regrouping, which is

Configuration	nDCG@10	ROI@10	PC@10	Balance	PC-lift@10
Random Forest baseline	0.019	+0.014	0.667	0.054	1.06×
LightGCN baseline	0.330	−0.005	0.794	0.428	1.17×
Segmentation	0.332	−0.007	0.821	0.411	1.19×
Segmentation + margin (model lever)	0.309	−0.007	0.918	0.407	1.40×
Segmentation + regrouping (data lever)	0.352	−0.006	0.949	**0.452**	n/a
Segmentation + regrouping + margin (combined)	0.333	−0.005	0.982	0.444	n/a

Table 3: Aggregate metrics averaged over the 69 temporal splits.

Band	$\pi(b)$	RF	LightGCN	Segment.	+Margin	+Regroup	Combined
Conservative	0.65	0.52	0.75	0.91	1.19	1.21	1.25
Income	0.78	0.70	1.13	1.20	1.26	1.25	1.27
Balanced	0.76	1.03	1.17	1.22	1.28	1.24	1.30
Aggressive	0.35	1.88	1.35	1.17	1.97	2.24	2.68

Table 4: Per-band PC-lift@10 (PC@$10/\pi(b)$) for the baselines and the segmentation configurations.

only possible if the declared bands had been adding noise to training. The declared bands, not the model, were holding performance back. The recommended operating point is therefore the data lever alone, since once the bands are corrected the margin loss is no longer necessary and even lowers the balance from 0.452 to 0.444.

6.3 Per-Band Suitability

The per-band lifts in Table 4 trace where each lever spends its effort:

- **Margin loss** is largely spent rescuing the two extremes the baselines failed, dragging Conservative from 0.75 past the random baseline to 1.19 and Aggressive from 1.35 to 1.97.
- **Segmentation alone** barely moves Conservative ($0.75 \rightarrow 0.91$, still below random) and even lowers Aggressive to 1.17, because partitioning by declared band still trains each sub-model on the same discordant purchases: the underlying labels stay uncorrected, so coherence cannot rise much.
- **Regrouping** clears the random baseline in every band (1.21, 1.25, 1.24, 2.24).
- **Combined** reaches the highest per-band lifts of all (1.25, 1.27, 1.30, 2.68).

The Conservative deficit that both baselines exhibited is thus closed by either lever, but only regrouping closes it without the accuracy penalty seen in the aggregate.

6.4 Consistency Across Splits

The aggregate gains are not artefacts of averaging over a few strong splits. Figure 6 gives the per-split deltas against the baseline and Figure 7 the win rate, the fraction of splits on which a configuration beats the baseline:

- PC@10 is positive on every one of the 69 splits for all three configurations.
- The balance delta is positive only for regrouping (+0.019 on average) and negative for both segmentation alone and the margin loss.
- Regrouping wins the balance on 71% of splits and nDCG@10 on all of them.
- The margin loss wins the balance on only 26% of splits and nDCG@10 on a mere 7%, so its accuracy cost is a uniform downward shift rather than a handful of bad splits.

The individual splits therefore tell the same story as the averages: regrouping beats the baseline, and the margin loss falls short of it.

7 Limitations and Future Work

Beyond the data caveats already noted for the analytical study (Section 3.6), several limitations bound the experimental results, each pointing to a direction for future work.

- *Single operating point and backbone.* We audit each baseline only at its primary-metric optimum (nDCG@10 for LightGCN, ROI@10 for Random Forest), report a single operating point for the margin lever on the inherited best-trial backbone hyperparameters rather than re-tuning end to end, and extend the segmentation only to LightGCN. The reported suitability gains are therefore a lower bound and the accuracy costs an upper bound. Applying the segmentation and its two levers to sequential, transformer, and content-based recommenders, and expanding the single margin point into a full Pareto sweep with end-to-end re-tuning, would map the balance frontier rather than report a single point on it.
- *Regrouping leakage.* The revealed bands are derived from each customer's full buy history, so they carry look-ahead information relative to each split. A per-split rolling variant using only training-window buys would remove this leakage.
- *Associational coherence premium.* The coherence premium of Section 3 is associational, not causal. Instrumental-variable or difference-in-differences designs on data spanning a regulatory boundary would move it towards causation, and would let the elevated

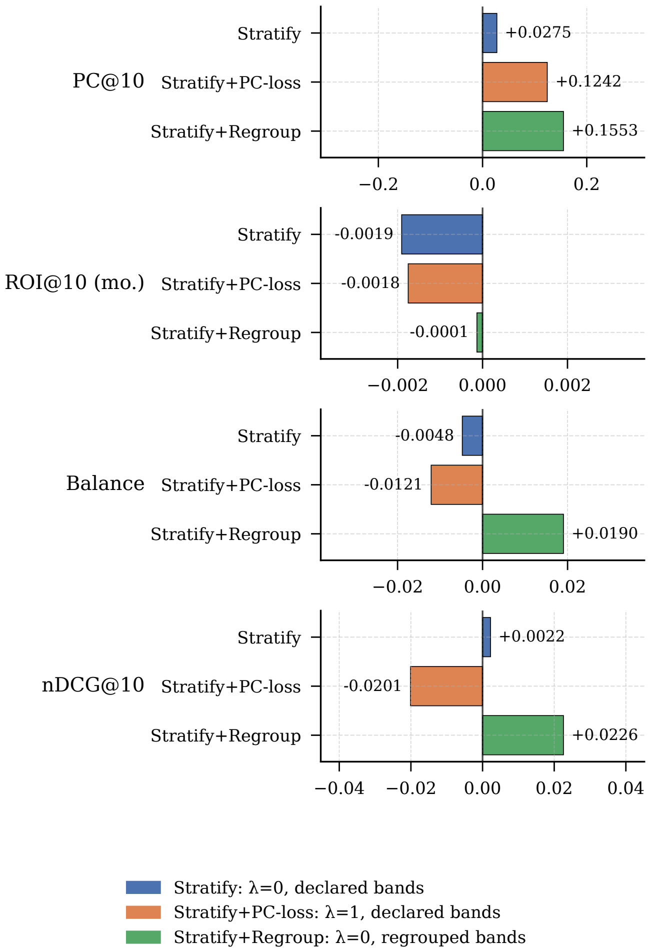

Figure 6: Paired per-split deltas against the LightGCN baseline.

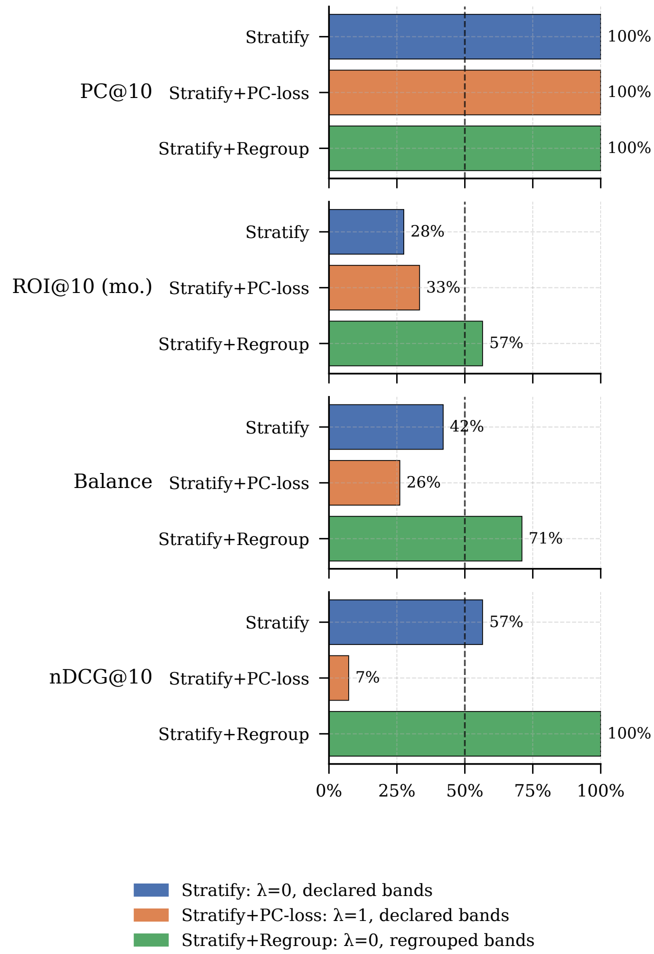

Figure 7: Per-split win rates against the LightGCN baseline.

2018 mean discordance be tested directly against a pre-MiFID II counterfactual rather than left as a hypothesised transition effect.

- *Hand-set metric parameters.* The ± 1 tolerance and the hierarchical asset-band assignment are set rather than derived. Calibrating the tolerance from data and validating the band assignment against external MiFID II classifications from fund providers would put both on firmer footing.

8 Conclusion

Our central finding is that return, ranking accuracy, and regulatory risk suitability need not conflict on FAR-Trans: a recommender can be made far more suitable to the customer's declared risk band while keeping, and even improving, accuracy and return. This makes suitability an architectural choice rather than a tradeoff patched by post-hoc re-ranking.

The results assemble a diagnostic-to-remedy chain. An analytical study establishes that discordance is prevalent and a persistent customer-level trait, and that coherent buys earn a statistically significant return premium, removing the economic objection to optimising for suitability. Auditing the FAR-Trans baselines on the three axes shows that both underserve specific risk bands and that neither balances the three. Risk-based segmentation then closes the per-band deficit. Of its two exclusive levers, regrouping customers onto their revealed band is the dominant one: it lifts the harmonic-mean balance above the baseline by raising suitability and accuracy together at no cost to return. The margin loss, by contrast, buys suitability by spending accuracy, and so lowers the balance.

More broadly, the approach shows that regulatory suitability constraints can be encoded as training-time objectives and, better still, corrected at the data source rather than filtered afterwards which opens a path for recommenders in other regulated domains, such as insurance product suitability.

9 Reproducibility

All code, configuration grids, and evaluation artefacts are available in the project repository at https://github.com/Samthesimpsons/Risk-Based-Segmentation-for-FAR.

References

[CNBC, 2019] CNBC. The stock market boomed in 2019. here's how it happened, December 2019. Accessed: 2026-06-08.

[Ghiye *et al.*, 2024] Ashraf Ghiye, Baptiste Barreau, Laurent Carlier, and Michalis Vazirgiannis. Rolling forward: Enhancing lightgcn with causal graph convolution for credit bond recommendation. In *Proceedings of the 5th ACM International Conference on AI in Finance*, ICAIF '24, pages 231–238. ACM, November 2024.

[He *et al.*, 2020] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. Lightgcn: Simplifying and powering graph convolution network for recommendation, 2020.

[Rendle *et al.*, 2012] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. Bpr: Bayesian personalized ranking from implicit feedback, 2012.

[Reuters, 2020] Reuters. S&p 500 completes recovery from covid-19 sell-off, hits record high, August 2020. Accessed: 2026-06-08.

[Reuters, 2021] Reuters. With fed taper expected, investors brace for rate hikes on horizon, November 2021. Accessed: 2026-06-08.

[Reuters, 2022] Reuters. Global markets wrap-up, December 2022. Accessed: 2026-06-08.

[Sakurai *et al.*, 2026] Keigo Sakurai, Takahiro Ogawa, Miki Haseyama, Anjyu Anan, and Kei Nakagawa. Risk-aware utility re-ranking for financial asset recommendation. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, WSDM '26, pages 1227–1231, New York, NY, USA, 2026. Association for Computing Machinery.

[Sanz-Cruzado *et al.*, 2024] Javier Sanz-Cruzado, Nikolaos Droukas, and Richard McCreadie. Far-trans: An investment dataset for financial asset recommendation, 2024.

[Sanz-Cruzado *et al.*, 2026] Javier Sanz-Cruzado, Richard McCreadie, Nikolaos Droukas, Craig Macdonald, and Iadh Ounis. Investors are (not) always right: A comparison of transaction-based and profitability-based metrics for financial asset recommendations. *ACM Trans. Inf. Syst.*, 44(2), January 2026.